Proof-governed autonomy for live revenue experiments

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Autonomous agents can write, route, and execute workflows, but commercial autonomy is hard to evaluate when constraints, buyer signals, and outcomes live in scattered logs. We report Relay, a continuously updated deployment record for a notes-first revenue agent governed by an explicit proof contract. The system converts a broad commercial goal into a state machine with permitted actions, forbidden actions, bottlenecks, proof windows, buyer-intent signals, payment outcomes, and compact research-journal entries. At the latest production read, the seven-day public-intent window contained 50 page views and 6 sample clicks, while the outbound proof remained in `stalled_opportunity_direct` with 9/10 active-sample sends, 0 replies, 0 payments, and USD 0.00 gross revenue. The result is therefore an execution finding, not a demand finding. The contribution is not a claim that Relay makes money. It is a falsifiable protocol for making autonomous revenue experimentation auditable enough that future success, failure, and regressions can be interpreted.

1 Introduction

Autonomous software agents are moving from demonstrations into deployed settings where they can observe, decide, and act through external tools (*1–5*). This creates a new evaluation problem. A tool-using system can appear autonomous because it sends messages, records data, or triggers payments, but a scientific claim of autonomy requires more than action. The system must preserve the state that justified action, the constraints that limited action, and the evidence that should update control.

This requirement is familiar in closed-loop experimentation. In autonomic computing, a controller monitors, analyses, plans, and executes under an explicit knowledge model (*6*). In closed-loop materials discovery, a system chooses the next experiment, records the result, and updates the search path (*7*). Commercial agent deployments need a related discipline, but with different risks: buyer privacy, outreach ethics, payment accounting, internal-test contamination, and human operator intervention. Without an explicit proof contract, a founder can repeatedly change the market, copy, price, or volume and still describe the whole process as one autonomous system.

Relay was built around a deliberately narrow offer: a buyer sends messy sales-call notes and receives a client-ready follow-up packet. The user-facing product is simple, but the research object is broader. Relay connects a product surface, buyer-intent telemetry, payment links, intake capture, outbound acquisition records, production-event routes, a success governor, an autonomous money loop, and a research-journal endpoint. The system is designed to keep the operator out of routine replanning during a proof window while preserving enough evidence to reconstruct what happened. Paper-backed changes are now treated as part of the evidence system: when the dashboard, backend, report language, source graph, metrics, guardrails, or results change materially, the manuscript and public PDF must be updated in the same work cycle.

Here we frame Relay as a proof-governed autonomy system. The term denotes a closed loop in which the agent can act only under a recorded proof state: a measurable objective, a current bottleneck, a permitted action, a forbidden-action set, a deadline, and outcome fields that define whether the action succeeded. This paper preserves the initial evidence lock and adds dated operating updates instead of replacing weak results with smoother language. The resulting claim is intentionally bounded. Relay is deployed and auditable; it is not yet proven profitable.

2 Results

The Results are organised around four levels of the deployment: the proof-governance protocol, the deployed product surface, the evidence matrix at the lock, and the journaled active experiment that determines the next interpretable claim.

2.1 A proof contract for autonomous revenue experiments

Relay converts an open-ended goal—make the system generate money while the operator stays out of the loop—into a constrained experiment. The proof contract contains five parts. First, a revenue objective defines the target outcome. Second, the state layer records money, payments, queue state, active-sample progress, send capacity, exception state, and recent market signals. Third, the success governor computes the bottleneck and selects the next allowed action. Fourth, the guardrail contract blocks actions that would make the experiment uninterpretable. Fifth, the research journal records the control tick and money-loop tick.

At the evidence lock, the success state was *active_experiment_sample*. The owner mode was *out_of_loop_waiting*. The allowed action was to send 10 active leads during the approved business-hour window. The forbidden set included asking the owner to choose copy, targeting, price, or volume before the proof deadline, increasing volume before buyer signal, changing more than one major variable at a time, and declaring demand failure from an execution miss. This converts an emotionally loaded commercial objective into a narrow experimental state.

Table 1. Proof contract at the evidence lock. The table separates the current state from the permitted action and the claims that remain unsupported.

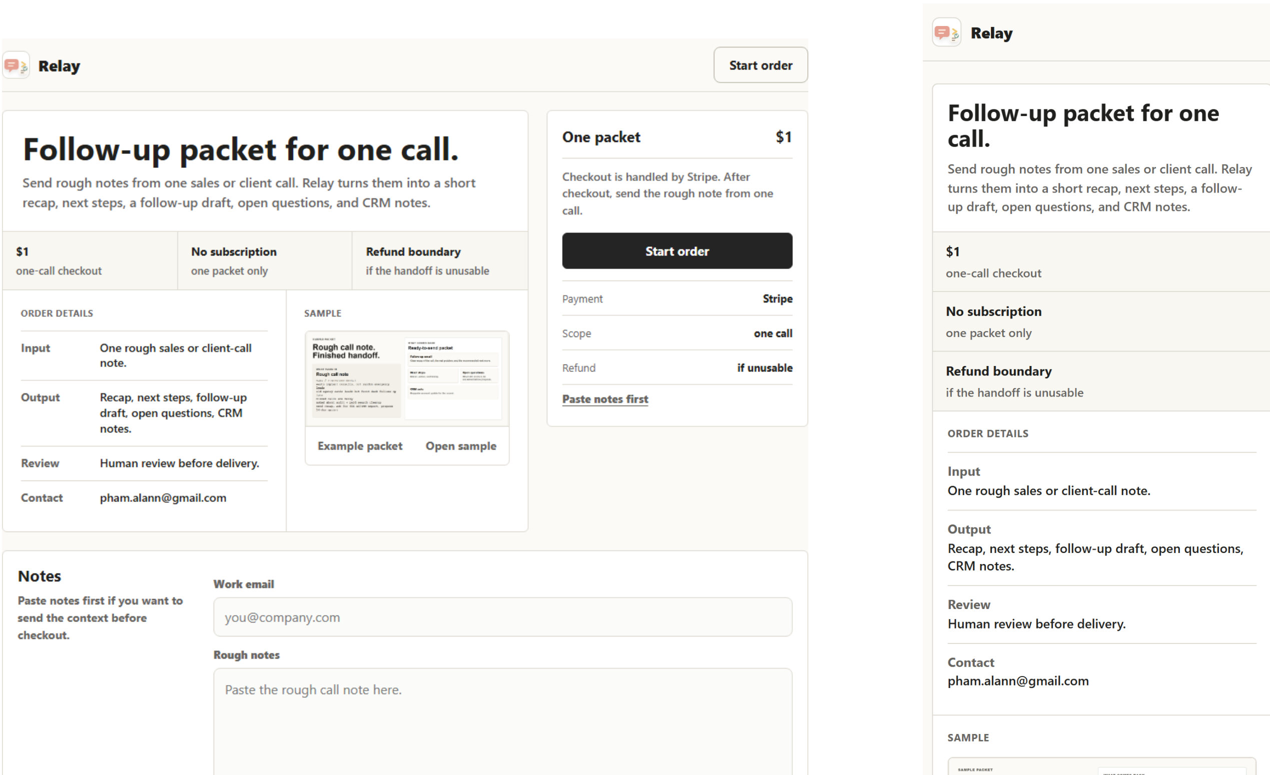
Element	Recorded value	Experimental function
Objective	Reach the first paid packet and a USD 1/week minimum-money proof	Defines the target outcome without changing the evidence.
Money state	USD 0.00 gross revenue; zero non-test payments	Prevents internal or unmatched test events from becoming a false success.
Bottleneck	Active hard-paid-test sample incomplete	Blocks premature judgement of demand.
Allowed action	Send 10 active leads inside the approved cap and window	Completes the minimum exposure before interpreting signal.
Forbidden actions	No owner-choice request, no volume increase, no multi-variable change, no failure claim from execution miss	Preserves causal interpretability.
Audit point	2026-05-04T17:00:00-04:00	Defines when the window should be evaluated.

2.2 The product surface is part of the experiment

The buyer-facing surface is intentionally direct. The current `relaybrief.com` page presents Relay as a standalone paid product rather than as an AO Labs project route: the header contains only the Relay identity and a neutral order action. A visitor can start a USD 1 one-call order, can use a lower-priority notes form if the notes are already available, and can view a sample that shows a rough note converted into a finished follow-up packet. These interface elements are not only conversion elements; they are also sensors. Viewing the sample, focusing the note field, submitting notes, clicking checkout, and entering an email all create evidence used by the proof contract.

Figure 1 shows the current production product surface used by the experiment. The desktop page presents the offer, USD 1 checkout, order terms, sample packet preview, and note-intake form in one view. The mobile view compresses the same path into a single-column flow. The preview is included here because future claims about buyer behaviour depend on what buyers actually saw.

The returned artifact is a follow-up packet. Figure 2 shows the sample preview used on the product page. This matters because the experimental offer is not a vague artificial-intelligence claim; it is a concrete promise that rough sales-call notes can be converted into a recap, follow-up draft, next-step table, open questions, and CRM-ready update.



(a) Desktop product page. **(b)** Mobile product page.

Figure 1. Buyer-facing Relay surface used by the evidence-locked deployment. The surface is both a product page and an experimental instrument: note-intake focus, sample clicks, lead capture, and checkout intent are recorded as buyer-intent signals.

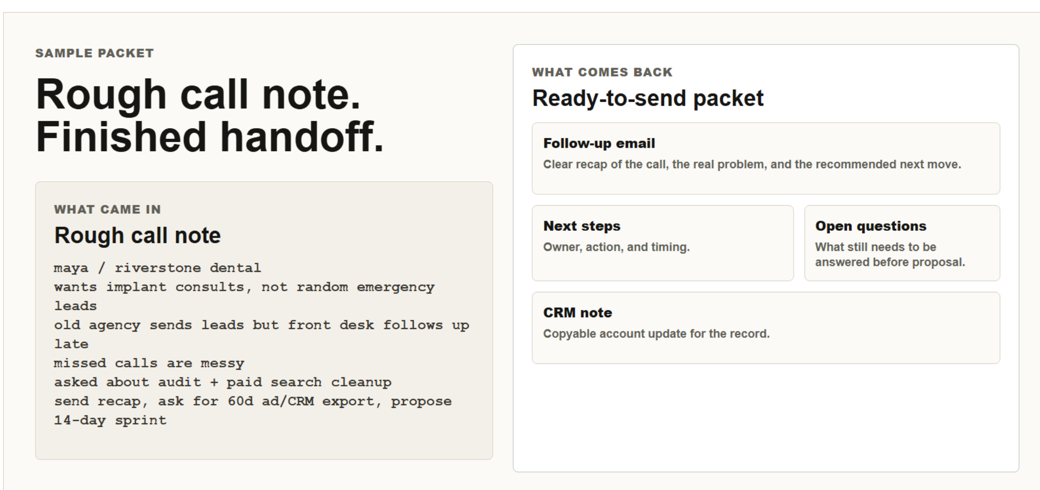


Figure 2. Relay sample-packet preview shown to buyers. The revised sample makes the value proposition visible as a before–after transformation: rough call note received, finished handoff returned.

2.3 Evidence at the lock supports deployment, not profit

The saved operational snapshot showed a live route surface and a configured backend, but the money state was still zero. Table 2 summarises the supported and unsupported claims. This distinction is the main scientific discipline of the manuscript. A stronger paper does not inflate early signals; it records where the proof stops.

Table 2. Evidence matrix at the 2026-05-04 lock. Values are descriptive and should not be read as statistical evidence of demand.

Domain	Evidence at lock	Supported claim	Unsupported claim
Route surface	Stripe and Tally webhooks, client gate, production routes, autopilot routes, intent summary, research journal	Relay was a deployed operational system, not only a static page.	Route robustness under buyer load.
Configuration	11 of 11 checked environment requirements present	Payment, intake, delivery, and tracking routes had required configuration.	Correct behaviour for every future edge case.
Intent	56 page views, 8 link clicks, 3 checkout clicks, 3 lead captures	There was measurable site interaction.	Product-market fit or statistically reliable conversion.
Filtered buyer signal	18 real page views, 1 real lead, 2 internal test leads separated	Internal activity was not treated as buyer demand.	A large external buyer sample.
Outbound	60 rolling seven-day sends, 0 replies, 0 payments, 0 send failures	Prior outreach and non-response were visible.	Outbound conversion.
Money	USD 0.00 gross revenue, zero non-test payments	Profit was not yet proven.	Autonomous money-making success.
Active experiment	0/10 active sends, 23 active due leads, 10/day cap	The next proof action was defined.	Demand failure before the sample completes.
Journal	4 compact research-journal entries	Control states were preserved for audit.	Longitudinal generalisation.

2.4 The active experiment defines the next interpretable result

At the evidence lock, the active experiment was a hard paid test. It had 0/10 progress, 10 sends remaining, 45 queued direct leads, 23 active due leads, and a remaining cap of 10. The next autonomous send window was 2026-05-04T09:30:00-04:00, and the proof audit point was 2026-05-04T17:00:00-04:00. Because the snapshot was taken before the send window, zero active sends was an expected pre-window state rather than evidence of market rejection.

The primary endpoint for the next window is non-test paid purchase count and gross revenue. Secondary endpoints are replies, checkout clicks, note submissions, lead captures, send failures, and open exceptions. If the active sample completes and no buyer signal appears, the protocol should rotate one major variable—targeting, copy, price, or volume—rather than rewriting the entire system. If buyer signal appears, the protocol should preserve the lane and close the payment or fulfillment loop.

Table 3. Predeclared interpretation of the next active sample. The goal is not to make a dashboard feel optimistic, but to make the next result interpretable.

Observed outcome	Interpretation	Permitted operation
Paid test or strong buyer reply	The lane has buyer signal.	Fulfil, preserve the lane, and continue only within the current cap.
Checkout clicks without payment	Friction may be in price, trust, or checkout path.	Change one variable and run another small sample.
No replies or clicks after complete sample	Current lane has no signal at this sample size.	Rotate one major variable; do not change targeting, copy, price, and volume simultaneously.
Send-window failure	Execution failed before demand was tested.	Fix execution before drawing market conclusions.
Manual owner intervention	Experiment becomes contaminated.	Mark the window as interrupted and separate it from autonomous evidence.

2.5 Current operating state

The latest live check was read from the production Relay API at approximately 2026-05-15T21:18Z. It updates the paper beyond the initial May 4 evidence lock without changing the interpretation standard. The system has more observed intent than the initial snapshot, but it still has no real buyer revenue. The previous `hard_paid_test_direct` reply-observation window matured with zero replies and zero payments, and the controller rotated without operator choice into `stalled_opportunity_direct`. The new send window produced a short execution result rather than a market result: the active sample reached 9/10, the current cap is zero, and no fresh first-touch active lead is queued for the final sample slot. The current source revision treats that as an active-experiment refill bottleneck, reserves only the missing active-sample slot, and allows remaining cap to keep direct-buyer follow-ups moving while fresh first-touch refill recovers.

Table 4. Current Relay state from the live production API. Values are operational state, not a hiring, market, or profitability probability.

Field	Live value
One-day intent	12 page views, 28 page-engagement events, 1 sample click, 0 leads.
Seven-day intent	50 page views, 94 page-engagement events, 6 sample clicks, 0 leads.
Revenue	USD 0.00 gross revenue; 0 payments.
Outbound execution	The 2026-05-15 send window produced 9/10 active-sample progress; the final active sample slot still requires fresh first-touch active leads and send capacity.
Active experiment	`stalled_opportunity_direct`; 9/10 active-sample sends observed; 1 active send remaining; 0 queued active first-touch leads for the remaining sample slot.
Controller state	`refill_direct_buyer_leads`; the live state marks the window blocked because the active experiment needs more queued first-touch leads and has no estimated send-window capacity.
First-money gates	The USD 1 checkout, product route, checkout-first page, single checkout path, notes fallback, telemetry, sample inspection, paid-fulfillment check, and reply-window protection are present; the real paid-buyer gate remains missing.
Public offer path	Landing page, sample PDF, checkout path, notes form, lead API, payment-webhook preflight, reply-autoclose preflight, and outbound preflight were reported as configured or OK.

2.6 A send-day accounting fault became part of the evidence

The May 11 operating read exposed an execution-accounting fault rather than a market conclusion. The controller showed `paid_test_explicit` at 10/20 active progress with ten queued active leads and no replies or payments. After UTC midnight, however, the live state also reported `outbound_window_missed` because the daily send counter could reset on the UTC day instead of the configured Eastern send day. The correct interpretation is not that the active lane failed; the sample had not completed, and the post-window miss state could be a time-accounting artefact.

The send-day accounting was revised so daily send counts begin at midnight in the configured send timezone rather than midnight UTC. The active paid-test path was also revised to expose a direct USD 1 first-money checkout, make the checkout-first route the first-screen action, add a source-visible primary-money-path marker, and preserve a lower-priority `#send-notes` fallback without presenting it as a send-first path. Later revisions made the reply-observation state authoritative across all money-control summaries. The public dashboard was then reduced to a compact revenue ledger: revenue, payments, checkout price, checkout clicks, the required first-payment condition, and a plain no-revenue status line, with raw data collapsed behind disclosure. A subsequent dashboard revision added a single cumulative sample-click line chart so the operator can see buyer inspection accumulating over time without restoring broad metric-card clutter. The buyer page was then separated from AO Labs suite navigation and rebuilt as a standalone `relaybrief.com` checkout surface with the USD 1 action as the only header action. The sample packet was then replaced with a before–after buyer proof showing the rough input, follow-up email, next steps, open questions, and CRM note. The latest source revision narrows the cap-reservation guard observed on 2026-05-15: if an active proof sample is incomplete and no active first-touch candidate is queued, the system reserves only the missing active-sample slots and lets direct-buyer follow-ups use the remaining cap. This update is recorded because an autonomous money-making system must learn from readiness, buyer-friction, state-consistency, and execution-accounting failures, not only from buyer outcomes.

2.7 Report language is part of the control surface

On 2026-05-06, the operator-facing Relay report language was changed from directive best-move framing to current-state framing in the autonomous operations summaries and related close-path/proposal text. This is not cosmetic. A system-owned operation should report state and current control logic without making the human operator read a task list for actions that the system should already own. The change is recorded here because paper-backed operational systems must keep their manuscripts aligned with meaningful dashboard, backend, and report-surface changes.

2.8 Compact journaling makes the system publishable as a research object

Before the journal layer, Relay could be monitored, but later reconstruction required multiple routes and human memory. The research-journal endpoint changes the status of the deployment because the system now records compact control entries. These entries preserve the money state, active-sample progress, proof state, health state, and current operation.

Table 5. Latest compact research-journal entries at the evidence lock. Buyer personal data are excluded from the manuscript.

Created at	Summary
2026-05-04T04:42:05.206321	money_loop money=$0.00 payments=0 sent=0 active_delta=0/0 active=0/10 proof=not_expected next=send 10 active leads and move progress from 0/10 to 10/10
2026-05-04T04:41:57.336593	success_control money=$0.00 payments=0 bottleneck=active_experiment_sample proof=prove_active_sample health=waiting_for_proof_deadline conversion_sent=0 failures=0
2026-05-04T04:40:17.309858	money_loop money=$0.00 payments=0 sent=0 active_delta=0/0 active=0/10 proof=not_expected next=send 10 active leads and move progress from 0/10 to 10/10
2026-05-04T04:40:14.073885	success_control money=$0.00 payments=0 bottleneck=active_experiment_sample proof=prove_active_sample health=waiting_for_proof_deadline conversion_sent=0 failures=0

3 Discussion

Relay is not yet evidence that an autonomous agent can make money without an operator. It is evidence that an autonomous commercial experiment can be constrained and logged so that a later money claim would be auditable. That is the stronger contribution. The system's most important result is the discipline of refusing to count an internal test as revenue, refusing to judge demand before the active sample and reply-observation window mature, refusing to change many variables at once, and separating buyer-signal failure from execution failure.

The broader implication is that agentic systems need proof contracts when deployed in domains with real-world consequences. A proof contract makes autonomy less theatrical and more scientific. It identifies what the system may do, what it must not do, when the result can be judged, and what evidence changes the next state. This structure could apply beyond Relay to other live agent deployments where market response, human intervention, or hidden tool actions make claims difficult to interpret.

The current deployment remains preliminary. It is a single system, a single offer, and a small descriptive dataset. It contains zero real revenue at the initial evidence lock and zero real revenue in the latest live check. It does not provide inferential statistics, external replication, or a mature ethics protocol. A Nature-level version of this work would need repeated journaled rotations after matured observation windows, non-test buyer outcomes, a comparison with manual or dashboard-only operation, a privacy-preserving dataset, independent review of code and logs, and a clear institutional position on commercial outreach and data retention.

4 Methods

4.1 Evidence lock and data sources

The initial manuscript lock used three saved production snapshots: `relay_ops_snapshot_2026-05-04T0042.json`, `relay_journal_snapshot_2026-05-04T0042.json`, and `relay_intent_snapshot_2026-05-04T0042.json`. The code state used for the initial evidence lock was commit `be6d9104aebd10dcf42b03d54f2d0ce1b8a9043b` in repository `nalalalan/relay-app`. The current-state update was read from live intent, ops, and journal routes in the production API. Raw snapshots and live responses are retained locally or in production systems for audit but are not reproduced in the public manuscript because they may contain operationally sensitive fields or personal identifiers.

4.2 Revenue accounting

Revenue was read from the money-system summary and success-controller state rather than from raw acquisition status counts alone. This prevents internal or unmatched Stripe test events from being counted as real buyer revenue. At the evidence lock, the money-system summary reported USD 0.00 gross revenue and zero payments. That value is used throughout the manuscript.

4.3 Intent and internal-test filtering

Intent was reported in two forms: all recorded intent events and filtered real-event counts where available. This is necessary because early deployment testing can otherwise mimic buyer demand. The manuscript reports aggregate counts only and excludes buyer emails and session identifiers.

4.4 Autonomy policy

The autonomy policy was read from the success-controller and money-system summaries. The policy defines the current bottleneck, allowed operation, forbidden actions, proof deadline, owner-interruption state, and recovery conditions. These values are treated as experimental controls. A future protocol deviation should be recorded in the journal before the corresponding outcome is interpreted.

4.5 Paper continuity

Relay is now treated as a paper-backed operational system. Any substantive change to the dashboard, backend, report language, source graph, evidence model, metrics, guardrails, or results must update the manuscript source and public PDF in the same work cycle. A paper-backed change is incomplete until the PDF rebuilds and the public artifact exposes the revised record.

4.6 Statistics and sample size

No hypothesis test was performed. The initial evidence-lock observations were descriptive: 56 page views, 18 filtered real page views, 8 link clicks, 3 checkout clicks, 3 lead-capture events, 1 filtered real lead, 60 rolling seven-day outbound sends, 0 replies, 0 payments, and 0 send failures. The latest live operating observations are also descriptive: 50 seven-day page views, 6 sample clicks, 0 leads, 0 replies, 0 payments, and USD 0.00 gross revenue. The active outbound state is `stalled_opportunity_direct` at 9/10 progress with an active-experiment refill bottleneck, so the current observation is a short active-sample execution fault rather than evidence of demand failure. The dashboard readiness state is an implementation-readiness measure, not a revenue result; the only revenue endpoint remains a non-owner Stripe payment. Future statistical claims require completed samples, matured observation windows, and a predeclared analysis plan.

4.7 AI assistance disclosure

This manuscript draft was prepared with OpenAI Codex under author direction using saved production snapshots and cited sources. The language model is not an author. The human author is responsible for factual verification, final approval, authorship decisions, and disclosure according to the target journal policy (*8*).

5 Data availability

The minimum dataset for this draft consists of the three saved JSON snapshots named in Methods plus source code needed to reproduce the route summaries and compact journal entries. Before public release, buyer emails, session identifiers, secrets, and operationally sensitive fields should be removed or transformed into privacy-preserving identifiers. A formal submission should deposit a frozen, sanitised dataset and schema in a DOI-minting repository or explain controlled-access restrictions (*9*).

6 Code availability

The implementation corresponds to repository `nalalalan/relay-app` at commit `be6d9104aebd10dcf42b03d54f2d0ce1b8a9043b`. Public release status must be confirmed before submission. If the code is central to the claims, peer reviewers should receive access to route definitions, the journal schema, a configuration template without secrets, and scripts or instructions needed to regenerate the manuscript figures from sanitised evidence.

7 Ethics and privacy

The system involves commercial outreach and buyer-intent logging. A complete study should specify lawful basis, retention period, opt-out handling, suppression-list behaviour, limits on automated follow-up, handling of personal data, and publication redaction rules. The present manuscript does not publish buyer personal data and separates internal tests from buyer signal.

8 Competing interests

The author owns or operates AO Labs and Relay and may benefit commercially if Relay succeeds. This commercial interest should be disclosed in any submission.

9 Funding

No specific funding source was recorded in the evidence used for this draft. The author should update this statement if the work falls within a grant, fellowship, institutional project, or company-funded effort.

10 Author contributions

A.N.P. conceived the system and research framing, directed the deployment goals, owns factual verification of the evidence, and is accountable for the final manuscript. AI assistance was used for drafting and formatting under author direction and is disclosed in Methods.

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